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### Cumulative localization architecture

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#### Abstract

A method for managing localized resources displayable during execution of a software product includes receiving a localization build request from a source control platform. The localization build request specifies a unique resource identifier and a resource value for each of multiple displayable resources associated with execution of the software product in a default geographic locale. For each displayable resource, the method further includes determining a relevant recency timestamp based on a build timestamp of the localization build request and retrieving, based on the relevant recency timestamp for the displayable resource, at least one translated resource indexed in association with the relevant recency timestamp. The method still further includes adding the at least one translated resource to a localization build for the software product and returning the localization build to the source control platform.

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## Background/Summary

### BACKGROUND

(1) Localization is the process of adapting content for presentation in a new region, including language translation, associated imagery, and/or cultural elements that influence how content will be perceived. Software products commonly provide users with the option to alter the display language for user interface elements displayed during execution of the code, such as text and/or graphic elements for buttons, menus, and other UI controls. Commonly, these types of displayable elements are developed, along with the source code, for presentation of GUI elements in a default geographic locale (e.g., an English-speaking locale). Following this initial development, the displayable elements are “localized” into translated counterparts configured for display in one or more other geographic locales. For example, English menu options such as “Save,” “Rename,” and “Print” are translated into Japanese, Chinese, French, German, etc.

(2) Currently, software development and localization are disconnected processes. A software production platform handles the default-language resources and code, while translation data is stored in an external localization database. The software production platform is a complex system

that uses version control and branching (e.g., development pertaining to different features of a same code product) to maintain aspects of software development, such as releases, product serving, feature development, and testing. One widely-used software production platform is Git, which supports both branching and version control extensively. This branching and version control functionality expedites and streamlines the development process by allowing teams developing different features of a same product and/or teams working on different versions of a same product to work within completely independent runtime environments.

(3) In contrast to software production, localization is typically managed by a single centralized localization database that does not support versioning or branching. When a displayable resource is updated within the localization database, the old version of the resource is replaced with the new one. One consequence of this common architecture is that existing localization databases can only link to one branch of a product at a time. Stated differently, any updates to a localization database have the potentially to affect (and disrupt) the work of all teams performing localization from the same database. This creates a plethora of challenges pertaining to management of localized resources for software products having different features and/or versions managed by different development teams. Current solutions are heavily reliant on manual intervention with database data, creation of database “snapshots” (e.g., copies of the database), and/or complicated and unnatural workarounds. These existing solutions are needlessly wasteful in terms of storage space utilization and unnecessarily complicated, reducing overall efficiency of the code production process.

#### SUMMARY

(4) According to one implementation, a localization system includes a cumulative localization database storing immutable time-stamped records corresponding to displayable resources accessed during execution of a software product. The system includes a resource record retriever that receives a unique resource identifier and a resource value for a displayable resource identified in a localization build request. In response, the resource record creator create a new resource record in cumulative localization database that includes the unique resource identifier, the resource value, and a build timestamp of the localization build request. The system further includes a translated resource retriever that requests a translation of the resource value corresponding to a locale specified in the localization build request. The translated resource retriever creates, in a translation table, a translation record that includes the translation, the unique resource identifier, and the build timestamp and adds the translated record to a localization build for the software product. The system outputs the localization build to a source control platform.

(5) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter.

(6) Other implementations are also described and recited herein.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A illustrates example aspects of a localization system having a traditional architecture.

(2) FIG. 1B illustrates example aspects of a localization system employing a cumulative localization database.

(3) FIG. 2 illustrates example actions for retrieving and creating localization data within a localization system including a cumulative localization database.

(4) FIG. 3 illustrates example operations of a localization system including a cumulative localization database.

(5) FIG. 4 illustrates further example operations of a localization system including a cumulative localization database.

(6) FIG. 5A illustrates example operations for performing a localization build utilizing a cumulative localization database.

(7) FIG. 5B illustrates additional example operations for performing a localization build utilizing a cumulative localization database.

(8) FIG. 6 illustrates an example schematic of a processing device **600** suitable for implementing aspects of the disclosed technology.

#### DETAILED DESCRIPTION

(9) The herein disclosed technology includes a localization architecture that provides significant technical improvement over previous approaches to localization in terms of cost-efficiency, scalability, and reduced complexity for improved user experience. The proposed architecture includes a localization database structured to handle the complex resource needs of software production, such as by allowing different product teams to work with different versions of the same localization resource without making local copies of the localization database (e.g., database snapshots captured at different times). The disclosed approach natively supports versioning, branches, and product servicing. Additionally, the disclosed localization architecture ensures data integrity by prohibiting editing of resources in the localization database and by preserving metadata on all changes accessible for inspection.

(10) FIG. 1A illustrates example aspects of a localization system having a traditional architecture and FIG. 1B illustrates functional aspects of a cumulative localization architecture employing the herein-disclosed technology. These two figures are intended to provide a high-level overview of the functional advantages that can be realized by implementing the disclosed cumulative localization architecture.

(11) The traditional architecture of FIG. 1A utilizes a localization system **102** with mutable records that change over time as updates are submitted. Data of the centralized localization database is pulled and utilized by different development teams working in parallel to perform different actions on source code being developed within a source control platform **104**. The source control platform **104** provides features that facilitate storing and tracking changes in association for different versions of a code product. The source control system **104** supports branching, meaning that developers can create separate development “branches” to effectively isolate code changes from other teams working on developing different portions of the same software product.

(12) The source control platform **104** illustrates an exemplary development pipeline that includes a development branch **116**, a release branch **118**, and various feature branches **120**. Circles on each of branches (e.g., branches **116**, **118**, **120**) represent different versions of a source code corresponding to the branch. For example, each circle on the development branch **116** represents a different version of a software product that underwent development, while each circle on the release branch **118** represents a different version of the software product that was eventually released. For example, circle **110** represents Microsoft Office 13, circle **112** represents Microsoft Office 14, circle **114** represents Microsoft Office 15, etc. Circles on the features branches **120** correspond to versions of different features within the code product. Different released version of the product (e.g., circles **110**, **112**, **114**) include updates to different product features. For example, the circle **122** represents a product feature updated in a first release version corresponding to the circle **112** (e.g., Microsoft Office 14), while the circle **124** represents a different product feature update that is in-development for a not-yet released version of the code product.

(13) The localization system **102** includes a localization database **132** that stores displayable resources. As used herein, a “displayable resource” refers to a data element that is called upon by source code and displayed to a user interface. For example, a displayable resource is a data element or collection of nested data elements that form some part of a user interface (e.g., an icon or selectable menu option including graphic elements and/or text). Displayable resources stored

within the localization database **132** are mutable in the sense that an update to any one of displayable resources replaces a previously-existing version of the same resource. In FIG. **1A**, database icons **126**, **128**, **130** represents modified versions of the localization database **132** at different points in time, where each modified version includes one or more new product resources or updates to previously-existing product resources that have replaced their former versions.

(14) Without complex and cost-burdensome work-arounds typically employed by enterprises using the architecture of the localization system **102** of FIG. **1A**, a modification to a displayable resource stored within the localization database **132** has the potential to undesirably impact many development teams at once. Assume, for example, a displayable resource corresponding to a UI menu item initially stores the string value “Home” and is subsequently changed to “Main Page.” Since the translation of “Home” into various languages may not mean the same as the literal translation of “Main Page” in those same languages, it is common practice to re-localize the resource and obtain new translations as part of each update. Since the strings “Home” and “Main Page” are of different lengths (and their respective translations may also be of different lengths), these updates have the potential to cause unexpected and/or undesirable UI changes.

(15) If all development teams for the code product were to always utilize the most current data in the localization database **132**, the above types of updates have the potential to undesirably impact many development teams at once, potentially creating small bugs that escalate into large and complex problems for lower-level teams working with code that inherits attributes of code managed by higher-level teams. A common work-around to prevent the foregoing entails creating different copies of the localization system **102** for each different team to work with. That is, each team grabs a complete copy, referred to as a “snapshot”, of the localization system **102** at a given point in time. The teams work with their respective snapshots-updating records and re-localizing records as needed.

(16) In addition to needlessly wasting space with duplicating thousands of records in multiple places, the above-described “snapshot” practice creates further complications stemming from the need to eventually synchronized all of the teams' updates to the localization database **132** prior to release of the code product. Consequently, the different product branches periodically synchronize their respective localization database changes. Synchronization is, in some systems, achieved by propagating a localization request from the requesting team to the root (top-level) development branch, performing localization within the localization system **102** (e.g., the main copy of the localization database), and creating a new snapshot of the localization system **102** that is, in turn propagated down to each development team. In a typical scenario, one of these snapshots includes hundreds of thousands of records. Replicating and transmitting this data repeatedly is time-consuming and wasteful in the sense that the multiple versions of the localized database (corresponding to different points in time) are concurrently stored and used, even in scenarios where only a handful of the stored resources are altered between snapshots.

(17) FIG. **1B** illustrates an alternative infrastructure incorporating an improved localization system **103** with a single localization database (referred to herein as a cumulative localization database) concurrently accessed and updated by many different development teams. The cumulative localization database store immutable localization records that facilitate synchronization of localized displayable resources with versioning of source code supported by the source control platform **104**. When a given displayable resource is modified within the localization database, the old version(s) of that resource are not modified or deleted. Rather, a new resource record is created for the resource, localization is performed to create corresponding new translations, and both the new resource record and the new corresponding translations are stored as immutable records for perpetuity. As used herein “immutable” means that the records are not, in the course of nominal database and/or localization system operations, changed or deleted.

(18) According to one implementation, each record in the cumulative localization database includes a timestamp and a resource identifier. The resource identifier uniquely identifies a resource but not

the resource content—that is, the resource identifier remains immutable as the content of the resource is updated. When a resource is updated, a new record is created for the resource. The new record includes the same resource identifier and a different (updated) timestamp along with other updated metadata discussed elsewhere herein. Each time a new resource record is created, localization is performed to create a set of translations corresponding to the updated resource content. Each translation is stored as a different record with a timestamp that matches the timestamp of the corresponding new resource record.

(19) One consequence of the above-described architecture (e.g., storing immutable records) is that each different branch of a code product can independently update resources stored within localization database without impacting previous records corresponding to the same resources. The timestamping of the different records effectively allows different teams to utilize select versions of the display resources that are guaranteed to remain unchanged.

(20) As shown in FIG. 1B, a first version of code on development branch **146** performs testing and development using a first set of localized displayable resources **150** while a feature branch **148** (e.g., a subbranch of the development branch **146**) performs testing and development using a second set of localized displayable resources **152**. The first set of localized displayable resources **150** and the second set of localized displayable resources **152** may be partially and/or largely overlapping (e.g., two hundred thousand identical records with fewer than 100 records that are modified); yet, all versions of the different respective resources and their respective localized (translated) counterparts are stored within the same localization database. This branch and version support offered by the improved localization system **103** allows development teams to work independently (e.g., without introducing database changes that impact other teams) without the complication and waste of creating a copy of the entire localization system for each different team to work with.

(21) By example, a released version of the code product **154** is packaged for customer distribution with a third set of resources **156**. The third set of resources **156** includes a complete collection of the resources utilized by the source code for the code product **154**. In the event that different feature branch teams built and tested GUI features using different versions of a same displayable resource, the third set of resources **156** packaged with the end product includes both versions of the resource for respective use by the corresponding branches in the source code.

(22) Beneficially, the above-described cumulative localization database architecture natively supports version control and branching that integrates with the standard software development process allowing each different development branches of a code product to utilize and independently update a respective subset of the resources in the localization database without impacting the work of other development teams building and testing other versions of the same product and/or different versions of different features of the product.

(23) The records in the cumulative localization database are immutable and have timestamps, which enable versioning to service the independent needs of different product branches. This eliminates the need for capture, distribution, and preservation of multiple “snapshots” of the localization database (many of which include high numbers of duplicative records), significantly reducing overhead pertaining to data storage and transfer. Additionally, since each branch can independently add records to the localization database and utilize select different versions of the same records, localization requests do not bottleneck the system (e.g., by causing all branches to wait for new data when a localization request is placed with the main copy of the localization system).

(24) FIG. 2 illustrates example actions for retrieving and creating localization data (e.g., creating a “localization build”) for a code product **206** within a localization system **200** that implements a cumulative localization database **214**. The code product **206** is developed within a source control platform **202** that supports version control and branching. In various implementations, the code product **206** is a stand-alone application or a subset of an application responsible for one or more

application features (e.g., a feature branch). When executed, the code product **206** generates and presents a graphical user interface (GUI) that serves content to a user including a variety of displayable resources. By example, a GUI menu **238** shows displayable resources **230, 232, 234** that each correspond to a different graphic and/or textual elements, such as drop-down menu option or selectable icon called upon during execution of the code product **206**. Although not shown, the displayable resources presented by the application may include some non-interactive displayable resources as well, such as instructional content (e.g., text files), template information, etc.

(25) The various displayable resources **230, 232, 234** of the code product **206** are defined, along with associated metadata, in one or more files, such as a displayable resource file **210**, which may be understood as storing at least a name defining each different displayable resource and a resource value for the displayable resource that includes the corresponding content populated on the GUI menu **228**. In one implementation, the value for each of the displayable resources **230, 232, 234** includes a text string. In another implementation, the value includes graphical information such as shape and color information, either alone or in addition to textual information. In various implementations, the displayable resource file **210** assumes different forms. By example, the displayable resource file **210** may be an XML or XML-like document that defines the displayable resources in a nested or tree-like structure.

(26) At one or more points in time during development of the code product **206**, a developer of the code product **206** request “localization” of the displayable resource file **210**. A used herein, localization refers to the process of adapting content for presentation in a new geographic locale different from the default geographic locale that the content is initially developed for presentation in. This adaptation typically includes translation of textual elements into one or more alternate languages.

(27) To request localization of the displayable resources defined within the displayable resource file **210**, the developer (or source control platform **202**) places a request with the localization system **216**. In the implementation shown, the displayable resource file **210** is submitted to the localization system **216** as an input of the request. The displayable resource file **210** is received by a parser that splits the raw data in the displayable resource file **210** into individual records that include a single displayable element and its associated metadata. Each of the individual record parsed from the displayable resource file includes at least a resource identifier as well as a value field defining the corresponding content that is displayed. The resource identifier uniquely identifies the resource by name and the value field defines the corresponding content that is displayed. Notably, the resource identifier remains fixed even while the value of the resource is modified.

(28) In addition to the above, each of the parsed records output by the parser **212** also includes a build timestamp. In one implementation, the build timestamp is sent by a client device as part of the localization request. The build timestamp may, for example, correspond to a time of the build request and/or time that is documented within the source control platform **202** in association with the code product **206**. For example, the build timestamp corresponds to a most recent update to the code product **206** in a version control system rather than the actual time of the build request.

(29) Depending upon the structure of the displayable resource file **210**, the parsed records may include other information as well. In one implementation, each of the parsed records includes a parent identifier that identifies another record containing the current record as a child in a tree structure. In another implementation, each of the parsed records includes a cargo field that contains all attributes of the resources and the value of the resource (e.g., a node within a tree-like structure).

(30) Each of the records parsed from the displayable resource file **210** is provided as input to a resource record retriever **218** that, in turn, queries a resource table **224** with the resource identifier and build timestamp of each parsed record. From this query, the resource record retriever **218** determines whether the cumulative localization database **214** already includes a record for the resource (e.g., a record with the same resource identifier). In one implementation, the resource record retriever **218** queries the resource table **224** for relevant records over a time interval

proceeding the build timestamp with an end that matches or that is otherwise determined based on the build timestamp each of the parsed records.

(31) If the resource table **224** does not already include a record associated with the resource identifier, a record creator **219** creates a new record in the resource table **224** based on the parsed record and assigns the record a timestamp that equal to or otherwise based on the build timestamp of the localization request. This timestamp is used as the “relevant recency timestamp” for the resource—the purpose of which is discussed further below.

(32) If, on the other hand, the resource table **224** does already include one or more records for the resource identifier, the resource record retriever **218** identifies and selects a newest-in-time in matching record that precedes the build timestamp. For this select record, the resource record retriever **218** determines whether value (e.g., resource content) of the existing record matches that of the parsed record. If the value of the parsed record does not match the select record, the record creator **218** creates a new record in the resource table **224** based on the parsed record and assigns the record a timestamp that equals or is otherwise based on the build timestamp, as described above. This timestamp is used as the “relevant recency timestamp” for the resource.

(33) If, on the other hand, the value of the parsed record does match the select record, no new record is created in the resource table **224** and the timestamp of the select record is used as the relevant recency timestamp for the resource.

(34) A translated resource retriever **222** performs actions for retrieving translated (localized) records corresponding to each of the parsed records. This translation retrieval and/or creation is performed based on the resource identifier and relevant recency timestamp of each of the parsed record (as set for the above) in addition to a set of requested locales that are, in one implementation, specified in the initial build request and/or in the displayable resource file **210**. The translated resource retriever **222** queries a translation table **226** with the resource identifier and relevant recency timestamp of each of the parsed records, as well as a designated list of locales of interest for the record. If the translated records for the requested locales are already stored in association with the relevant recency timestamp and resource identifier, these translated records are retrieved and added to a localization build **228** along with the corresponding parsed record that the translation query was based on (e.g., the English version of the record).

(35) If there is no translated record currently stored in association with the relevant recency timestamp and resource identifier for one or more of the requested locales, a resource translator **220** translates the parsed resource (e.g., employing various machine learning models and translation logic) for the request locale(s) to create a set of translation records that are then stored in the translation table **226** in association with the relevant recency timestamp and resource identifier of the corresponding parsed record. These newly-created translations are added to the localization build **228** along with the corresponding English-version record.

(36) The localization system **216** outputs the localization build **228**, which includes each of the resources in the displayable resource file **210** with values configured for output in a first default locale (e.g., an English-speaking locale) as well as corresponding translated values for these resources in each of the requested locales.

(37) FIG. 3 illustrates further example operations **300** of a localization system performed in response to receiving a localization build request. As used herein, a localization build request is a request placed—e.g., by a developer or source control platform—to generate a “localization build” that is, essentially, a bundle of displayable resources called upon by a source code product. A localization build typically includes a same set of displayable resources configured for display in at least two geographic locales. For example, localization build include strings that are in a language of a default geographic local (e.g., English-language strings) and the same strings translated into one or more additional languages.

(38) The localization build request specifies a displayable resource file **310** including raw data for each of various displayable resources called by a code product. For each of the displayable



resources, the displayable resource file **310** includes raw data identifying a corresponding unique resource identifier and a resource value that stores the presentable content of the resource in a format associated with presentation in a default locale (e.g., an English-speaking locale).

(39) In response to receipt of the localization build request at the localization system **300**, the displayable resource file **310** is provided as input to a parser **312**. The parser **312** reads the raw data of the displayable resource file **310** and splits this data into individual “records” (e.g., R1 data, R2 data, R3 data, each corresponding to a single displayable resource). Each of these records is populated with data parsed from the displayable resource file **310** and then input, sequentially, to a resource record retriever **306**.

(40) In FIG. **3**, a parsed record **323** illustrates exemplary information included in each record that is output by the parser **312**. This information includes a unique resource identifier **318**, a resource value **320**, a hash value **322**, a build timestamp **324**, and a list of locale(s) of interest **342**. Each of these fields is briefly described below.

(41) In general, the unique resource identifier **318** is an identifier that uniquely identifies the associated displayable resource within a source control platform. This identifier is, for example, unique within an extensive pool of displayable resources used across multiple teams working on different projects all managed within the source control platform also across different branches of the particular product that the displayable resource pertain to. By example, the unique resource identifier **318** is shown to include a project identifier **326**, a branch identifier **328**, and a resource name **330**. The project identifier **326** is an example of a namespace that may apply to a number of different teams working on different versions of a code product. For example, there may be several development teams working on features for the application “Microsoft PowerPoint” (identified by project id “ppt”). The branch identifier **328** is an example of an identifier associated with a given namespace (e.g., project id), that applies to team(s) working on code associated with a single version of the product (e.g., Microsoft Office15). Some resource identifiers may not include the branch identifier **328**. For example, the branch identifier **328** is excluded for some displayable resources developed at the root level for use used by many or all lower branches of a code product.

(42) The resource name **330** is a name that is assigned to the displayable resource, such as a developer-assigned description name or a default name generated by the source control platform at the time the displayable resource is initially defined.

(43) The resource value **320** stores the value of the associated displayable resource (the resource value). In the example shown, the value is a simple string with the English words “Save As.” In other examples and implementations, the value may be shorter, longer, or of other (e.g., non-string) forms.

(44) The hash value **322** is a mathematically-created unique identifier usable to identify the value of the associated record. For example, the hash value **322** is a hash encoding the resource value **320-a** string: “Save As.” The hash value **322** is generated via any known hashing methodology and is typically represented as a smaller number of total bits as compared to the corresponding value being represented.

(45) The build timestamp **324** of the parsed record **323** identifies a timestamp associated with the localization build request. In one implementation, the build timestamp **324** indicates a time that the localization build request is placed; in other implementations, the build timestamp **324** matches a stored timestamp corresponding a last update to a code product version in a version control system that is selectively presents the displayable resources defined within the displayable resource file **310**. In still another implementation, the build timestamp **324** is a timestamp that is explicitly specified in the build request, such as a timestamp corresponding to a time period of interest to a developer. In another implementation, the build timestamp **324** reflects a build time of the displayable resource file (e.g., a timestamp identifying when the displayable resource file **310** was created).

(46) The list of locales of interest **342** includes one or more geographical locales of interest that are

different from the default geographic locale of the displayable resource file **310**.

(47) For each record received as input, the resource record retriever **306** initiates a query to identify whether the identified record already exists within a resource table **327** of the cumulative localization database **314**. The resource table **327** is to be understood as comprising localization records configured for display in the default locale (e.g., an English-speaking locale). In the example shown, the resource record retriever **306** queries the resource table **327** based on the unique resource identifier **318** of the parsed record **323** and the build timestamp **324** of the parsed record **323**.

(48) In one implementation, the above-described query is formulated to return all records matching the unique resource identifier **318** and the resource value **320** of the parsed record **323** that are created during a time interval having an end that matches or that is otherwise defined based on the build timestamp **324** of the build request.

(49) In one implementation, the unique resource identifier of a stored record is identified as “matching” the unique resource identifier **318** of the parsed record **323** when either (1) all components of the unique resource identifier **318** match that of the stored record (e.g., project identifier, branch identifier, and resource name) OR (2) the project identifier and resource name of the unique resource identifier **318** match the stored record, and the stored record does not have a branch identifier (e.g., the branch ID is null).

(50) In the illustrated example, the build timestamp **324** is “6/15/2023 06:22” and the query returns all records matching the unique resource identifier **318** that were created at or prior to this time. The resource table **327** stores two records **332**, **334** indexed in association with the same project id and resource name as the parsed record **323**. Notably, the record **332** is indexed in association with a different branch identifier **344** (“Office 14”) than the branch identifier **328** in the parsed record **323**. Therefore, the record **332** is not returned in the query. However, the record **334** is returned since it includes the same resource name (“idsResave”), project identifier (“ppt”), and branch identifier (“Office15”) as the parsed record **323**.

(51) In this example, the record **334** is the sole matching record returned in the query and is therefore identified as the relevant default-locale record for the displayable resource. In some implementations, the resource record retriever **306** performs additional operations to verify that the value of the parsed record **323** matches the value of the stored record. For example, the hash value **322** of the parsed record **323** is compared to a stored hash value **346** within the record **334**. Confirming that the hash values match ensures the existing record corresponds to a same version of the record. Still other implementations may utilize hashes to capture file metadata, such as a “comment hash” based on and usable to uniquely identify comments stored in association with a displayable resource. For example, the resource record retriever **306** verifies the value matches by ensuring ensure that both the hash value **322** and hash comment (not shown) fields of an identified existing table record match that in the corresponding parsed record **323**.

(52) Once a single record has been identified as the relevant default-locale record corresponding to the parsed record **323** (e.g., as described above), the resource record retriever **306** temporarily stores a timestamp of the relevant default-locale record (e.g., “Jun. 13, 2023 20:34”) as the “relevant recency timestamp **336**” for the associated displayable resource. The relevant recency timestamp **336** is key in subsequent operations pertaining to obtaining corresponding translations, as is discussed further below.

(53) In some queries, the unique resource identifier **318** of the parsed record **323** corresponds to two or more existing records in the resource table **327**. This is the case when the value of the displayable resource is updated by a given team. For example, a “PowerPoint” development team working on version “Office15” updates the resource value **320** of the displayable resource from “ReName” to “Save As” and, in doing so, creates a new record with an updated timestamp in the resource table **327** without deleting an older version of the record. In this scenario where multiple records match the unique resource identifier **318**, the timestamps in the resource table **327** are used

to identify the relevant default-locale record from among those matching the unique resource identifier **318**. For example, the resource record retriever **306** identifies a newest-in-time record that precedes that of the build timestamp **324** as the relevant default-locale record. The hash value of the this is, in some implementations, compared to the hash value **322** of the parsed record **323** to further confirm accuracy of the match.

(54) If the resource table **327** does not store any records matching both the unique resource identifier **318** and the resource value **320** of the parsed record **323**, record creator **319** creates a new record in the resource table **327** that includes data of the parsed record **323** that has been parsed from the displayable resource file **310**. This newly-created record is stored in the resource table **327** with a timestamp that matches the build timestamp **324** (or that is otherwise based on the build timestamp **324**). The resource record retriever **306** uses this new record as the relevant default-locale record and temporarily stores the build timestamp **324** as the relevant recency timestamp **336** for the displayable resource.

(55) The identified or newly-created relevant default-locale record corresponding to each parsed record is added to a localization build **348**, which is to be understood as including a bundle of displayable resources called upon by the associated code product.

(56) In summary, the resource record retriever **306** identifies a relevant default-locale record for each parsed record that it receives as input. This relevant default-locale record is either a recycled (matching) existing record or a newly created record. The timestamp of this relevant default-locale record is temporarily stored (as the relevant recency timestamp **336**) and used to obtain a set of translated records corresponding to the relevant default-locale record.

(57) In one implementation, the relevant recency timestamp **336** for each displayable resource is passed, along with the unique resource identifier **318** and the locales of interest **342** to a translated resource retriever **338**. The translated resource retriever **338** identifies or creates (if not already existing) a corresponding set of translated resources **340**.

(58) FIG. **4** illustrates further example operations **400** of a localization system performed in response to receiving a localization build request. Specifically, FIG. **4** illustrates a translated resource retriever **402** and a resource translator **404** that perform actions for obtaining translated records (e.g., newly-created records or recycled existing records) corresponding to each displayable resource named in a localization build request (e.g., within the displayable resource file **310** of FIG. **3**). In one implementation, the localization system **400** further includes some or all components shown and described with respect to either of FIG. **2** and FIG. **3**.

(59) As input, the translated resource retriever **402** receives data of a default locale record **406** extracted from a resource table (not shown) of a cumulative localization database **414**. The default locale record **406** represents the “relevant record” that is identified for given displayable resource via the operations described with respect to FIG. **3**.

(60) In FIG. **4**, the default locale record **406** includes a unique resource identifier **418**, a relevant recency timestamp **436**, a list of locales of interest **440**, a resource value **420**, and a hash value **422** that encodes the resource value **420**. In one implementation, the unique resource identifier **418** includes the same or similar subcomponents as those described with respect to the unique resource identifier **318** of FIG. **3**. The relevant recency timestamp **436** is a timestamp for a corresponding relevant record identified as described above with respect to FIG. **3**. The list of locales of interest **440** identifies a one or more geographical locales that the displayable resource is to be configured for presentation in (e.g., locales specified as inputs to the localization build request). The resource value **420** includes displayable content associated with a default geographic locale (e.g., an English-language string associated with an English-speaking locale). In FIG. **4**, the annotation “EnHash Value” is used to denote the fact that the hash value **322** encodes the English-language value of the resource, as compared to other hashes (hash value **460**) included in a translation table **426** that encode a corresponding translated value (e.g., value **462**) of the resource.

(61) Each record stored in the translation table has a timestamp **442** that is, upon creation of the

record, set to match the relevant recency timestamp **436** of the corresponding relevant record (e.g., default locale record) that the translation is derived from.

(62) The translated resource retriever **402** queries a translation table **412** with some or all of the corresponding default locale record **406**. For example, the query specifies at least the unique resource identifier **418**, the relevant recency timestamp **436**, and the locale(s) of interest **440**. In response to this query, the translated resource retriever **412** receives records from the translation table **426** that match the specified inputs.

(63) In one implementation, an existing translated record is “recycled”—meaning, added to a localization build **428**—when both the unique resource identifier **418** and the relevant recency timestamp **436** of the default locale record **406** match that of the stored translated record for one of the specified locales of interest **440**. If both of these fields do not match exactly, a resource translator **404** creates and stores a new translated record based on the default locale record **406**, such as by using translation tools and methods readily available in the art. This newly-created record is added to the translation table **426** and the timestamp **442** for the new record is set to equal that of the relevant recency timestamp **436** for the record of interest.

(64) Due to the systematic matching of the timestamp **442** of each translated record to the relevant recency timestamp **436** of the corresponding default-locale record (e.g., a record storing the English version of the same resource), the relevant recency timestamp **436** is a key piece of information that can be leveraged to determine whether a translated record is an accurate translation of the value stored in the default locale record **406**. When a translated record in the translation table **426** includes a timestamp and a unique resource identifier that match the relevant recency timestamp **436** and the unique resource identifier **418** in the default locale record **406**, this means that the translated record was derived from the default locale record **406**.

(65) In some implementations, the translated resource retriever **412** performs additional verification of a translated record by confirming that the hash value **422** (e.g., EnHash Value) of the default locale record **406** matches that of a corresponding hash value **430** of the translated record. This match confirmation further guarantees, with absolutely certainty, that the stored translated record is derived from the default locale record **406**.

(66) For each of the locale(s) of interest **440** included the default locale record **406**, a matching translated record is either recycled or newly-created as described above. These matching records are then added to the localization build **428**. After the localization build **428** is updated to include the default-locale version and corresponding translations of interest for each resource identified the localization build request, the localization build **428** is output to a source control platform. The localization build **428** is, in some implementations, added to a downloadable package provided to an end user in association with the software product. For example, the localization builds created by each of multiple different teams can be merged and deduplicated into a final localization build (and/or a final localization build can be run for the entire product at once, specifying multiple different displayable resource files as input for each of multiple branches of the corresponding code product). The localization build for the final version of the software product is either distributed with the software product or made available for download from a centralized (e.g., online) repository. For example, the code product is configured to download the appropriate localization build during initial installation or nominal execution on a user machine.

(67) Notably, the localization build **428** is branch specific (e.g., for different versions of a code product and/or different features of the code product), meaning that different development teams can actively add resources to and pull resources from the cumulative localization database **414** without altering resources that the other development team(s) are relying on. This eliminates the need to have redundant “snapshot” copies of the localization database available for each different team to work with in isolation. The above is realized, in part, by the usage of the herein-described unique resource identifiers in combination with the disclosed timestamping scheme that (1) that ensures each default-locale record has a timestamp that is based on one specified in a

corresponding initial build request; and (2) each translated record has a timestamp that matches the timestamp of the default-locale record that it is derived from.

(68) FIG. 5A illustrates example operations **500** for performing a localization build utilizing a cumulative localization database. A receiving operation **502** receives a build request that identifies, for each of multiple displayable resources, a resource identifier and a resource value. A selection operation **504** selects a displayable resource from the multiple displayable resources. In some implementations, the selection operation **504** entails parsing a displayable resource file into individual parsed records that each correspond to one of the displayable resources identified in the displayable resource file (e.g., as described with respect to FIG. 3). A first one of the parsed records is selected.

(69) A querying operation **506** queries a resource table of the cumulative localization database based on the unique resource identifier of the selectable displayable resource and a build timestamp associated with the build request. In various implementations, the build timestamp corresponds to a timestamp of the build request, a versioning timestamp associated with particular version of source code associated with the requested build, a creation timestamp of a displayable resource file provided as input to the build request, a developer-specified input to the build request, or other timestamp, such as the further example build timestamps discussed herein with respect to FIG. 3.

(70) A matching operation **508** determines whether the resource table includes one or more default-locale records that match the criteria specified by the querying operation **506**. For example the matching operation **508** determines whether the resource table stores one or more existing records that have the same unique resource identifier and a timestamp that precedes or matches the build timestamp associated with the build request. If the matching operation **508** determines that one or more matching default-locale records do exist, a selection operation **510** selects, from the matching records, a newest-in-time record that precedes or matches the timestamp of the build request.

Following the selection operation **510**, a verification operation **512** confirms whether or not a resource value of the selected record matches the resource value of the displayable record.

(71) If not, a determination operation **514** determines whether there exist other matching records and, if so, a selection operation **516** selects a next newest-in-time matching record that precedes the timestamp of the build request. In this flow, the verification operation **512**, determination operation **514**, and selection operation **516** are repeated as described above until the verification operation **512** confirms a match between the resource value of the selected record and the resource value of the displayable record.

(72) Once the verification operation **512** confirms that the stored value of the selected record matches the resource value of the displayable record, a storing operation **518** stores the record as the default-locale record for the displayable resource.

(73) If, on the other hand, either (1) the verification operation **512** fails to confirm a match of the displayable resource's resource value with a stored value of any record returned in the query or (2) the matching operation **508** finds that the resource table does not store any default-locale records matching the initial criteria specified by the querying operation **506**, a record creation operation **520** creates a stores a new record in the resource table. This new record includes a timestamp that matches or that is otherwise based on (derivable from) the build timestamp of the build request. This newly-created record is identified as the default-locale record for the displayable resource.

(74) A retrieval operation **522** retrieves a timestamp of the default-locale record and uses this timestamp as the “relevant recency timestamp” for the displayable resource. Following the retrieval operation **522** the operations **500** proceed to operation **524** of FIG. 5B.

(75) FIG. 5B illustrates further example operations **501** for performing a localization build utilizing a cumulative localization database. In one implementation, the example operations **501** are performed following the example operations **500** described with respect to FIG. 5A.

(76) The operations **501** being with a querying operation **524** that queries a translation table of the cumulative localization database based on the unique resource identifier and the relevant recency

timestamp for the default-locale record corresponding to the selected displayable resource. For example, the querying operation requests records in the translation table that have the same unique resource identifier as the default-locale record and a timestamp that matches the relevant recency timestamp of the default-locale record.

(77) A matching operation **526** determines whether matching records are returned by the query operation. In one implementation, a returned record is identified as “matching” the criteria of the query operation **524** if the record stores a hash value encoding the resource value of the default-locale record. In scenarios where one or more matching translations records are identified, an identification operation **528** then identifies a subset of the identified matching translation records that correspond to locale(s) specified in the build request. A build operation **530** adds these records to the localization build.

(78) A determination operation **532** next determines whether translation records have been acquired for all locales of interest identified in the build request. If either (1) a translation record is not-yet-acquired for one or more of the locales or (2) the matching operation **526** does not identify any matches to the criteria specified in the query operation **524**, a translation request operation **534** requests, from a translation source, a translation of the resource value for each of the not-yet-acquired locales of interest specified in the build request. A storing operation **536** stores each of the newly-generated translations as a different translation record including both (1) a timestamp that matches the relevant recency timestamp of the default-locale record and (2) a resource identifier that matches the unique resource identifier of the default-locale record. In some implementation, the newly-created translation records each additionally store a hash value encoding the resource value that was translated in order to create the record. The build operation **520** adds these newly-generated translation records to the localization build.

(79) A determination operation **538** determines whether the build request identified other displayable resources not yet added to the localization build. If so, the operations **501** proceed to back to the selection operation **504** of FIG. 5A. If not, an output operation **540** outputs the localization build.

(80) FIG. 6 illustrates an example schematic of a processing device **600** suitable for implementing aspects of the disclosed technology. The processing device **600** includes one or more processor unit(s) **602**, memory device(s) **604**, a display **606**, and other interfaces **608** (e.g., buttons). The processor unit(s) **602** may each include one or more CPUs, GPUs, etc.

(81) The memory device(s) **604** generally includes both volatile memory (e.g., RAM) and non-volatile memory (e.g., flash memory). An operating system **610**, such as the Microsoft Windows® operating system, the Microsoft Windows® Phone operating system or a specific operating system designed for a gaming device, resides in the memory device(s) **604** and is executable by the processor unit(s) **602**, although it should be understood that other operating systems may be employed.

(82) One or more applications **612** (e.g., the improved localization system **103** and/or the source control platform **104** of FIG. 1B) are loaded in the memory device(s) **604** and executed on the operating system **610** by the processor unit(s) **602**. The applications **612** may receive inputs from one another as well as from various input local devices such as a microphone **634**, input accessory **635** (e.g., keypad, mouse, stylus, touchpad, gamepad, racing wheel, joystick), and a camera **632**. Additionally, the applications **612** may receive input from one or more remote devices, such as remotely-located smart devices, by communicating with such devices over a wired or wireless network using more communication transceivers **630** and an antenna **638** to provide network connectivity (e.g., a mobile phone network, Wi-Fi®, Bluetooth®). The processing device **600** may also include one or more storage devices **628** (e.g., non-volatile storage). Other configurations may also be employed.

(83) The processing device **600** further includes a power supply **616**, which is powered by one or more batteries or other power sources and which provides power to other components of the

processing device **600**. The power supply **616** may also be connected to an external power source (not shown) that overrides or recharges the built-in batteries or other power sources.

(84) The processing device **600** may include a variety of tangible computer-readable storage media and intangible computer-readable communication signals. Tangible computer-readable storage can be embodied by any available media that can be accessed by the processing device **600** and includes both volatile and nonvolatile storage media, removable and non-removable storage media. Tangible computer-readable storage media excludes intangible and transitory communications signals and includes volatile and nonvolatile, removable and non-removable storage media implemented in any method or technology for storage of information such as computer readable instructions, data structures, program modules or other data. Tangible computer-readable storage media includes RAM, ROM, EEPROM, flash memory or other memory technology, CDROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other tangible medium which can be used to store the desired information, and which can be accessed by the processing device **600**. In contrast to tangible computer-readable storage media, intangible computer-readable communication signals may embody computer readable instructions, data structures, program modules or other data resident in a modulated data signal, such as a carrier wave or other signal transport mechanism. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, intangible communication signals include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

(85) Some implementations may comprise an article of manufacture. An article of manufacture may comprise a tangible storage medium (a memory device) to store logic. Examples of a storage medium may include one or more types of processor-readable storage media capable of storing electronic data, including volatile memory or non-volatile memory, removable or non-removable memory, erasable or non-erasable memory, writeable or re-writeable memory, and so forth. Examples of the logic may include various software elements, such as software components, programs, applications, computer programs, application programs, system programs, machine programs, operating system software, middleware, firmware, software modules, routines, subroutines, operation segments, methods, procedures, software interfaces, application program interfaces (API), instruction sets, computing code, computer code, code segments, computer code segments, words, values, symbols, or any combination thereof. In one implementation, for example, an article of manufacture may store executable computer program instructions that, when executed by a computer, cause the computer to perform methods and/or operations in accordance with the described implementations. The executable computer program instructions may include any suitable type of code, such as source code, compiled code, interpreted code, executable code, static code, dynamic code, and the like. The executable computer program instructions may be implemented according to a predefined computer language, manner or syntax, for instructing a computer to perform a certain operation segment. The instructions may be implemented using any suitable high-level, low-level, object-oriented, visual, compiled and/or interpreted programming language.

(86) In some aspects, the techniques described herein relate to a system for managing localized resources displayable during execution of a software product. The system includes: a cumulative localization database storing immutable time-stamped records corresponding to displayable resources accessed during execution of a software product. The system further includes a resource record retriever stored in memory that is executable to receive, for a displayable resource identified in a localization build request, a unique resource identifier and a resource value and to create an immutable record in the cumulative localization database that includes the unique resource identifier, the resource value, and a build timestamp of the localization build request. The system still further includes a translated resource retriever stored in memory and executable to: request a

translation of the resource value corresponding to a geographic locale specified in the localization build request; create, within a translation table of the cumulative localization database, a translation record that includes the translation, the unique resource identifier, and the build timestamp; and add the translated record to a localization build for the software product, wherein the system outputs the localization build to a source control platform.

(87) In some aspects, the techniques described herein relate to a system wherein the unique resource identifier includes a resource name, a project identifier, and a branch identifier.

(88) In some aspects, the techniques described herein relate to a system, wherein the build timestamp corresponds to a versioning timestamp associated with a version of source code for the software product stored within the source control platform.

(89) In some aspects, the techniques described herein relate to a system, wherein the resource record retriever is further configured to: query a resource table of the cumulative localization database for records within a time interval having an end defined based on the build timestamp of the localization build request; and in response to determining that the resource table does not include a record indexed in association with the unique resource identifier, create the translation record.

(90) In some aspects, the techniques described herein relate to a system, wherein the displayable resource is one of multiple displayable resources identified within a data file provided as an input to the localization build request and the system is further configured to create a separate default-locale record in the cumulative localization database for each of the multiple displayable resources identified within the data file.

(91) In some aspects, the techniques described herein relate to a system, wherein the resource table stores immutable records associated with a default locale, each of the immutable records corresponding to a displayable resource that is presented to a user display during execution of the software product.

(92) In some aspects, the techniques described herein relate to a method including: receiving a localization build request from a source control platform, the localization build request specifying a unique resource identifier and a value for each of multiple displayable resources associated with execution of a software product in a default geographic locale; for each displayable resource of the multiple displayable resources: determining a relevant recency timestamp for the displayable resource based on a build timestamp of the localization build request, the relevant recency timestamp for the displayable resource corresponding to an immutable record stored within a localization database; retrieving, based on the relevant recency timestamp for the displayable resource, at least one translated resource indexed in association with the relevant recency timestamp; adding the at least one translated resource to a localization build for the software product; and returning the localization build to the source control platform, the localization build being added to a downloadable package provided to an end user in association with the software product.

(93) In some aspects, the techniques described herein relate to a method, wherein the unique resource identifier includes a resource name, a project identifier, and a branch identifier.

(94) In some aspects, the techniques described herein relate to a method, wherein determining the relevant recency timestamp of the displayable resource further includes: determining a timestamp corresponding to a newest-in-time record indexed in association with the unique resource identifier and created prior to the build timestamp of the localization build request.

(95) In some aspects, the techniques described herein relate to a method, wherein determining the relevant recency timestamp of the displayable resource further includes: querying a resource table of the localization database based on the unique resource identifier, the value of the displayable resource, and the build timestamp of the localization build request; and in response to querying the resource table, receiving one or more records from the localization database, wherein the relevant recency timestamp matches a timestamp of a newest-in-time record selected from the one or more



records.

(96) In some aspects, the techniques described herein relate to a method, further includes: querying a resource table of the localization database based on the unique resource identifier, the value of the displayable resource, and the build timestamp of the localization build request; and in response to determining that the resource table does not include a resource record indexed in association with the unique resource identifier and the value of the displayable resource, creating a new record in the resource table for the displayable resource, wherein the relevant recency timestamp matches a build timestamp of the localization build request.

(97) In some aspects, the techniques described herein relate to a method, wherein retrieving the at least one translated resource further includes: querying a translation table based on the unique resource identifier and the relevant recency timestamp of the displayable resource.

(98) In some aspects, the techniques described herein relate to a method, wherein the translation table stores immutable translation records.

(99) In some aspects, the techniques described herein relate to a method, wherein the method further includes requesting generation of the at least one translated resource in response to determining that the translation table does not include a record matching the unique resource identifier and the relevant recency timestamp.

(100) In some aspects, the techniques described herein relate to a method, wherein the localization build request specifies a plurality of geographic locales and requesting generation of the at least one translated resource further includes: obtaining at least one translated record from the translation table indexed in association with the unique resource identifier and the relevant recency timestamp for the displayable resource; adding the at least one translated record to the localization build.

(101) In some aspects, the techniques described herein relate to one or more tangible computer-readable storage media encoding computer-executable instructions for executing a computer process including: receiving a localization build request from a source control platform, the localization build request specifying a unique resource identifier and a value for a displayable resource, the displayable resource being associated with execution of a software product in a default geographic locale; determining a relevant recency timestamp for the displayable resource based on a build timestamp of the localization build request, the relevant recency timestamp for the displayable resource corresponding to an immutable record stored within a localization database; retrieving, based on the relevant recency timestamp for the displayable resource, at least one translated resource corresponding to the displayable resource; adding the at least one translated resource to a localization build for the software product; and outputting the localization build to the source control platform, the localization build being added to a downloadable package provided to an end user in association with the software product.

(102) In some aspects, the techniques described herein relate to one or more tangible computer-readable storage media, wherein the unique resource identifier includes a resource name, a project identifier, and a branch identifier.

(103) In some aspects, the techniques described herein relate to one or more tangible computer-readable storage media, wherein determining the relevant recency timestamp of the displayable resource further includes: determining a timestamp corresponding to a newest-in-time indexed in association with the unique resource identifier and created prior to the build timestamp of the localization build request.

(104) In some aspects, the techniques described herein relate to one or more tangible computer-readable storage media, wherein determining the relevant recency timestamp of the displayable resource further includes: querying a resource table of the localization database based on the unique resource identifier and the build timestamp of the localization build request; in response to querying the resource table, receiving one or more records from the localization database and determining the relevant recency timestamp of the displayable resource based on a record timestamp of a newest-in-time record selected from the one or more records.

(105) In some aspects, the techniques described herein relate to one or more tangible computer-readable storage media, wherein the computer process further includes: in response to determining that the resource table does not include a record indexed in association with the unique resource identifier, creating a new record in the resource table for the displayable resource and determining the relevant recency timestamp for the displayable resource based on the build timestamp of the localization build request; requesting generation of at least one translated resource based on the value of the displayable resource; and indexing at least one translated resource in association with the unique resource identifier and the relevant recency timestamp in a translation table.

(106) The logical operations described herein are implemented as logical steps in one or more computer systems. The logical operations may be implemented (1) as a sequence of processor-implemented steps executing in one or more computer systems and (2) as interconnected machine or circuit modules within one or more computer systems. The implementation is a matter of choice, dependent on the performance requirements of the computer system being utilized. Accordingly, the logical operations making up the implementations described herein are referred to variously as operations, steps, objects, or modules. Furthermore, it should be understood that logical operations may be performed in any order, unless explicitly claimed otherwise or a specific order is inherently necessitated by the claim language. The above specification, examples, and data, together with the attached appendices, provide a complete description of the structure and use of example implementations.

## Claims

1. A system for managing localized resources displayable during execution of a software product, the system including: memory; a processing system; a cumulative localization database stored in the memory, the cumulative localization database storing immutable time-stamped records corresponding to displayable resources accessed during execution of a software product; a resource record retriever stored in the memory and executable by the processing system to: receive, for a displayable resource identified in a localization build request, a unique resource identifier and a resource value, the displayable resource being one of multiple displayable resources identified within a data file provided as an input to the localization build request; and create an immutable record in the cumulative localization database for each different one of the multiple displayable resources identified within the data file, the immutable record for the displayable resource including the unique resource identifier, the resource value, and a build timestamp of the localization build request; a translated resource retriever stored in memory and executable to: request a translation of the resource value corresponding to a geographic locale specified in the localization build request; create, within a translation table of the cumulative localization database, a translation record that includes the translation, the unique resource identifier, and the build timestamp; and add the translation record to a localization build for the software product, wherein the system outputs the localization build to a source control platform.
2. The system of claim 1 wherein the unique resource identifier includes a resource name, a project identifier, and a branch identifier.
3. The system of claim 1, wherein the build timestamp corresponds to a versioning timestamp associated with a version of source code for the software product stored within the source control platform.
4. The system of claim 1, wherein the resource record retriever is further configured to: query a resource table of the cumulative localization database for records within a time interval having an end defined based on the build timestamp of the localization build request; and in response to determining that the resource table does not include a record indexed in association with the unique resource identifier, create the translation record.
5. The system of claim 4, wherein the resource table stores immutable records associated with a

default locale, each of the immutable records corresponding to a displayable resource that is presented to a user display during execution of the software product.

6. A method comprising: receiving a localization build request from a source control platform, the localization build request specifying a unique resource identifier and a value for each of multiple displayable resources associated with execution of a software product in a default geographic locale; for each displayable resource of the multiple displayable resources: querying a resource table of a localization database based on the unique resource identifier, the value of the displayable resource, and a build timestamp of the localization build request; receiving, in response to querying the resource table, one or more immutable records from the localization database; determining a relevant recency timestamp for the displayable resource, the relevant recency timestamp for the displayable resource matching a timestamp of a newest-in-time record selected from the one or more immutable records; retrieving, based on the relevant recency timestamp for the displayable resource, at least one translated resource indexed in association with the relevant recency timestamp; adding the at least one translated resource to a localization build for the software product; and returning the localization build to the source control platform, the localization build being added to a downloadable package provided to an end user in association with the software product.

7. The method of claim 6, wherein the unique resource identifier includes a resource name, a project identifier, and a branch identifier.

8. The method of claim 6, wherein determining the relevant recency timestamp of the displayable resource further comprises: determining a timestamp corresponding to a newest-in-time record indexed in association with the unique resource identifier and created prior to the build timestamp of the localization build request.

9. The method of claim 6, further comprises: querying a resource table of the localization database based on the unique resource identifier, the value of the displayable resource, and the build timestamp of the localization build request; and in response to determining that the resource table does not include a resource record indexed in association with the unique resource identifier and the value of the displayable resource, creating a new record in the resource table for the displayable resource, wherein the relevant recency timestamp matches a build timestamp of the localization build request.

10. The method of claim 7, wherein retrieving the at least one translated resource further includes: querying a translation table based on the unique resource identifier and the relevant recency timestamp of the displayable resource.

11. The method of claim 10, wherein the translation table stores immutable translation records.

12. The method of claim 10, wherein the method further includes requesting generation of the at least one translated resource in response to determining that the translation table does not include a record matching the unique resource identifier and the relevant recency timestamp.

13. The method of claim 10, wherein the localization build request specifies a plurality of geographic locales and requesting generation of the at least one translated resource further includes: obtaining at least one translated record from the translation table indexed in association with the unique resource identifier and the relevant recency timestamp for the displayable resource; adding the at least one translated record to the localization build.

14. One or more tangible computer-readable storage media encoding computer-executable instructions for executing a computer process comprising: receiving a localization build request from a source control platform, the localization build request specifying a unique resource identifier and a value for a displayable resource, the displayable resource being associated with execution of a software product in a default geographic locale; querying a resource table of a localization database based on the unique resource identifier and a build timestamp of the localization build request; in response to querying the resource table, receiving one or more records from the localization database; determining a relevant recency timestamp for the displayable

resource based on a record timestamp of a newest-in-time record selected from the one or more records, the relevant recency timestamp for the displayable resource corresponding to an immutable record stored within a localization database; retrieving, based on the relevant recency timestamp for the displayable resource, at least one translated resource corresponding to the displayable resource; adding the at least one translated resource to a localization build for the software product; and outputting the localization build to the source control platform, the localization build being added to a downloadable package provided to an end user in association with the software product.

15. The one or more tangible computer-readable storage media of claim 14, wherein the unique resource identifier includes a resource name, a project identifier, and a branch identifier.

16. The one or more tangible computer-readable storage media of claim 14, wherein determining the relevant recency timestamp of the displayable resource further comprises: determining a timestamp corresponding to a newest-in-time indexed in association with the unique resource identifier and created prior to the build timestamp of the localization build request.

17. The one or more tangible computer-readable storage media of claim 14, wherein the computer process further comprises: in response to determining that the resource table does not include a record indexed in association with the unique resource identifier, creating a new record in the resource table for the displayable resource and determining the relevant recency timestamp for the displayable resource based on the build timestamp of the localization build request; requesting generation of at least one translated resource based on the value of the displayable resource; and indexing at least one translated resource in association with the unique resource identifier and the relevant recency timestamp in a translation table.

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